# LearnSPH: A Tool for Teaching the Basics of Smoothed Particle Hydrodynamics in Graphics Research

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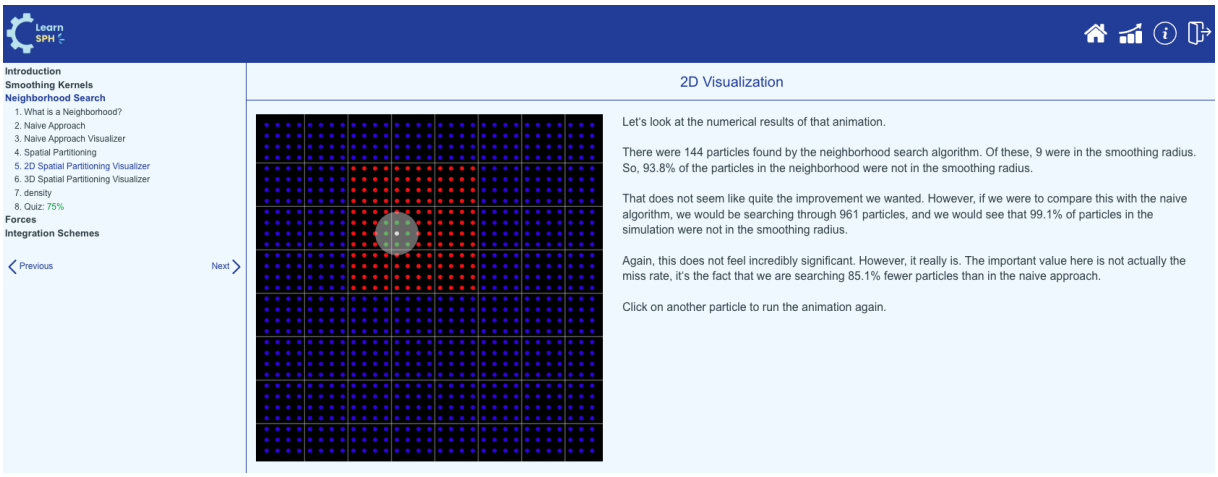



**Figure 1: LearnSPH's in-framework view with the hierarchical window of sections on the left and the content area on the right.**

## Abstract

Smoothed Particle Hydrodynamics (SPH) is a mesh-free Lagrangian method used to simulate continuum data, which was initially developed for astrophysical applications [5]. Over the past twenty years, SPH has been applied in various other areas, including Computer Graphics. The graphics community has always been interested in physically based simulations for achieving high-quality visual effects and imitating real-world phenomena. SPH has become one of these simulations that has attracted the attention of researchers. However, learning and using SPH can be challenging. Although SPH offers remarkable advantages, such as meshless simulation and suitability for complex geometries, its learning curve is very high. It requires careful implementation of boundary conditions and convergence, and incurs a significant computational cost.

Additionally, parameter tuning can be a challenge for users, particularly those new to the method. Although there are open-source SPH resources and various publications, nearly all of them are aligned with the mentioned high learning curve, and understanding the basics of the method remains challenging for beginners. Over the past couple of years, we have observed that interactive learning is crucial for student success, as it increases engagement and fosters critical thinking skills. Here, we propose an interactive web-based tool to teach the basics of SPH to those new to the field. We aim to provide them with a deeper understanding through visualized conceptual explanations and practical applications.

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## 1 Motivation

Particle-based fluid simulations have gained increased interest over the years, featuring in everything from movies and commercials to scientific simulations [4]. However, to master SPH, one must first overcome the substantial learning curve and have a strong understanding of smoothing kernels, physics, and many advanced topics in computer graphics. While academic papers and open-source engines provide valuable resources for beginners, they often assume prior expertise and fail to provide step-by-step explanations.

This project is intended for students, educators, and researchers interested in SPH. Prerequisites for using this platform are surface-level familiarity with graphics programming and calculus. Users with some prior experience in SPH can skip many foundational sections and go straight to sections on their specific weaknesses, appealing to users at all levels of SPH knowledge.

Upon completing the entire LearnSPH platform, a user would have the conceptual understanding needed to implement an SPH

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simulation, work with others on SPH projects, and understand the literature surrounding SPH, enabling them to stay up to date on the latest innovations in the field.

## 2 Our Contribution

We propose an interactive web-based learning platform that lowers the barrier to entry for SPH simulations by:

- Breaking down complex topics into short and clear lessons.
- Providing interactive visualizers to help users build an intuitive understanding of SPH topics.
- Including periodic quizzes for self-assessment and reinforcement.

We teach these concepts in four sections:

(1) **Basics:** Introduction, Smoothing Kernels, Neighborhood Search, Force Computations, and Integration Schemes
(2) **Numerical Aspects:** Handling Pressure Field and Introducing Solvers, Boundary Handling, Surface Tension, and Viscosity
(3) **Additional Details:** Rigid-Fluid coupling, Deformable-Fluid Coupling, Multi-Phase Fluids, and Data-Driven Models
(4) **Surface Generation:** Surface Tracking, Marching Cubes, Rendering, and Optimization

## 3 User Experience

Learning on SPH is done through a clear and structured path:

(1) **Explanation:** A user will read through short lessons that often feature equations, code snippets, or visuals to aid their understanding. By breaking down the complex topic of SPH into submodules and, within those submodules, into short lessons, we make the subject more approachable and improve retention. The strategy of breaking a complex topic down into short lessons has been proven effective in our goals of engagement and understanding [7].
(2) **Practice:** A user will apply these newly learned concepts through interactive visualizations to help them understand what is happening on a single simulation step. Visualization of algorithms has been shown to improve learning among engaged learners, and, as SPH is fundamentally algorithmic, the visualizations provided in LearnSPH should help users understand SPH [3].
(3) **Reinforcement:** A user will then be quizzed on this material. Our quizzes can be retaken at any time and show users which questions they got right and wrong, thus enabling spaced repetition and providing feedback that users can use to inform their studying and further solidify their understanding. Both of these techniques are known to improve long-term retention of information [6].

## 4 Implementation

The website itself is created using NextJS [8] with a PostgreSQL [2] database for storing user information and their quiz scores. Visualizations are created using ThreeJS [1], which allows for user interactivity in 2D and 3D. As user interaction on mobile is far less reliable than on desktop, we have made the website inaccessible on phones and tablets.

## 5 Educational Applications

LearnSPH can be used in both the education of researchers and professionals interested in SPH simulations, as well as in traditional lecture-based graphics classes to teach the basics to students. Those aiming to learn all the fundamentals surrounding SPH simulations will benefit from going through all the lessons, visualizations, and quizzes included in LearnSPH. In a classroom, visualizations and interactive sections can be used to assist a professor in explaining concepts that are not intuitive without visual aids, such as neighborhood search.

## 6 Future Work

In the near future, we plan to continue developing and testing LearnSPH. As of now, testing on LearnSPH has only been conducted by expert users; however, we plan to conduct a study upon completion of the whole platform. In this study, we will aim to answer the following questions:

(1) What is the rate of improvement in the learning process, and what are the performance-based outcomes for the first-time learners?
(2) How do visualizations, animations, and gamification affect the learning of beginners?

We will use surveys and exams given through the platform to a group of students studying the fields related to SPH to answer these questions.

Feedback from expert users, both educators and student researchers, who have used the platform, notes that the UI is clean and effective, and that the interactive lessons helped cover difficult-to-visualize topics such as neighborhood search.

## 7 Conclusion

LearnSPH aims to lower the barrier to entry for a particularly complicated subject in graphics education. Our combination of structured lessons, interactive practice sessions, and quizzes will bridge the gap between the theory behind SPH and an intuitive understanding of it.

## References

[1] Ricardo Cabello. 2010. *Three.JS.* https://threejs.org/
[2] PostgreSQL Global Development Group. 1986. *PostgreSQL.* https://www.postgresql.org
[3] Christopher D. Hundhausen, Sarah A. Douglas, and John T. Stasko. 2002. A Meta-Study of Algorithm Visualization Effectiveness. *Journal of Visual Languages Computing* 13, 3 (2002), 259–290. doi:10.1006/jvlc.2002.0237
[4] Dan Koschier, Jan Bender, Barbara Solenthaler, and Matthias Teschner. 2022. A Survey on SPH Methods in Computer Graphics. *Computer Graphics Forum* 41, 2 (2022), 737–760. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.14508 doi:10.1111/cgf.14508
[5] Matthias Müller, David Charypar, and Markus Gross. 2003. Particle-based fluid simulation for interactive applications. In *Proceedings of the 2003 ACM SIGGRAPH/Eurographics Symposium on Computer Animation* (San Diego, California) *(SCA '03)*. Eurographics Association, Goslar, DEU, 154–159.
[6] Henry L. Roediger and Andrew C. Butler. 2011. The critical role of retrieval practice in long-term retention. *Trends in Cognitive Sciences* 15, 1 (2011), 20–27. doi:10.1016/j.tics.2010.09.003
[7] Balasundaram Sathiyaseelan, Jain Mathew, and Sridevi Nair. 2024. Microlearning and Learning Performance in Higher Education: A Post-Test Control Group Study. *Journal of Learning for Development* 11 (03 2024), 1–14. doi:10.56059/jl4d.v11i1.752
[8] Vercel. 2016. *Next.js.* https://nextjs.org